

ChurnIQ

Bank Customer Churn Prediction — Complete Technical Report

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Category: Data Science / Machine Learning · Industry: Banking & Financial Services

Status: Live — model trained and deployed; documentation and validation ongoing

Live app: yemifatodu-bank-churn-predictor.streamlit.app · Repo: github.com/yemifatodu/bank-churn-predictor

1. Executive Summary

ChurnIQ is a customer attrition prediction system built for retail banking. It trains a Random Forest classifier on 10,000 historical customer records and exposes it through a live, deployed Streamlit application that returns a real-time churn risk verdict, a feature-driver breakdown, and a tailored retention action plan for any customer profile entered.

The model achieves 86.80% accuracy and a ROC-AUC of 0.864 on a held-out test set. The three strongest predictors of churn are customer age, number of products held, and account balance, together accounting for roughly two-thirds of the model's decision weight.

Honest caveat: overall accuracy is inflated by class imbalance (79.6% of customers do not churn). The model's recall on the churn class is 0.44 — it currently identifies fewer than half of customers who actually leave, while being highly precise (0.83) when it does flag someone. This is documented transparently throughout this report rather than obscured behind the headline accuracy figure, and is the primary target for the next iteration of work.

This report documents the dataset, methodology, exploratory findings, model development, evaluation, and business implications of the project as it currently stands, and distinguishes clearly between what has been built and validated versus what remains as future work.

2. Introduction

Customer churn — a customer closing their account or ceasing to use a bank's services — is a persistent and costly problem in retail banking. This project builds a supervised machine learning pipeline to predict, from a customer's demographic and account profile, the probability that they will churn, and packages that prediction into a usable decision-support tool rather than a static analysis.

3. Problem Statement

Retail banks typically learn about customer attrition only after an account has closed, at which point retention is no longer possible. Existing customer data — credit score, account balance, product holdings, tenure, activity level — contains predictive signal about attrition risk, but that signal is rarely converted into a forward-looking, actionable score that a relationship manager or retention team can use before a customer leaves.

4. Business Background

Customer acquisition in banking is materially more expensive than retention, and churned customers represent both lost revenue and lost cross-sell opportunity. A model that scores existing customers by churn risk allows a bank to target retention spend — personalized offers, proactive outreach, product incentives — at the customers most likely to leave, rather than applying blanket retention campaigns.

Specific quantified business impact (e.g. cost per acquisition, average customer lifetime value, projected ROI of a retention campaign informed by this model):

[Information Needed]

5. Objectives

Primary objective

Build a machine learning model that predicts the probability of an individual bank customer churning, using demographic and account attributes available at scoring time.

Secondary objectives

- Identify which customer attributes most strongly drive churn risk, to inform business strategy beyond the prediction itself.
- Package the model into an interactive tool usable by a non-technical stakeholder, not just a notebook.
- Evaluate the model honestly at the class level, not just on aggregate accuracy, given known class imbalance in the target.

Success criteria / KPIs

KPI	Type	Target / Result
Overall accuracy	Technical	86.80% (achieved)
ROC-AUC	Technical	0.864 (achieved)
Churn-class recall	Technical	0.44 (achieved — flagged as improvement target)
Business KPI: reduction in realized churn after deployment	Business	[Information Needed — no post-deployment measurement available yet]

6. Target Users

- Retention / relationship management teams — primary consumers of the risk score and action plan.
- Product and marketing teams — consumers of the feature-driver insights for strategy (pricing, bundling, engagement).
- Bank leadership — consumers of aggregate risk-segment reporting for budget allocation.

Specific named stakeholders, internal champions, or pilot users:

[Information Needed]

7. Dataset

Source and size

Bank Customer Segmentation dataset (based on the widely used Kaggle 'Bank Customer Churn' / Churn_Modelling.csv dataset), 10,000 rows, 13 columns.

Columns

Column	Type	Description
CustomerId	Integer	Unique customer identifier
Surname	String	Customer surname (not used as a model feature)
CreditScore	Integer	Customer credit score
Geography	String	Country: France, Germany, or Spain
Gender	String	Male or Female
Age	Integer	Customer age in years
Tenure	Integer	Years as a bank customer (0–10)
Balance	Float	Account balance
NumOfProducts	Integer	Number of bank products held (1–4)
HasCrCard	Integer	1 if customer holds a credit card, else 0
IsActiveMember	Integer	1 if customer is an active member, else 0

EstimatedSalary	Float	Estimated annual salary
Exited	Integer	Target — 1 if customer churned, else 0

Data quality

- Missing values: none (0 across all 10,000 rows and 13 columns).
- Duplicate rows: none. Duplicate CustomerId values: none.
- Class balance: 20.37% churned (2,037 customers) vs. 79.63% retained (7,963 customers) — a real, non-trivial class imbalance.

Potential bias and ethical considerations

The dataset covers only three countries (France, Germany, Spain), so the model's learned patterns — particularly the strong Germany effect described in the EDA section — reflect this specific population and should not be assumed to generalize to other markets without revalidation. Gender is used as a model feature; because Gender shows a real association with churn in this data (see Section 9), any deployment should include a fairness review to confirm the model is not creating disparate outcomes across gender groups in downstream retention actions.

8. Methodology

The project follows a standard supervised classification workflow: exploratory data analysis → preprocessing (scaling and encoding) → train/test split → model training → evaluation → deployment as an interactive application. The full pipeline, from raw CSV to trained model, is implemented in a single training script (`train.py`) that also exports the fitted model and preprocessor as artifacts consumed by the deployed app.

9. Data Cleaning

Because the source dataset has no missing values and no duplicate records, no imputation, deduplication, or outlier removal was required or performed. Preprocessing was limited to feature scaling and encoding (detailed in Section 11).

Outlier analysis (IQR method, from the project notebook): CreditScore - 15 outliers; Age - 359 outliers; NumOfProducts - 60 outliers; Tenure, Balance, and EstimatedSalary - 0 outliers. A defensive check for zero-value CreditScore or Age records found none. No outlier removal was performed - all records were retained for modeling.

10. Exploratory Data Analysis

Overall churn rate across the full dataset: 20.37%.

Summary statistics

Feature	Mean	Std Dev	Min	Max
CreditScore	650.5	96.7	350	850
Age	38.9	10.5	18	92
Balance	76,485.9	62,397.4	0	250,898.1

Churn rate by geography

Geography	Customers	Churn Rate
France	5,014	16.15%

Germany	2,509	32.44%
Spain	2,477	16.67%

Germany shows roughly double the churn rate of France or Spain, despite having a similar customer base size to Spain. This is the single largest geographic disparity in the data and warrants business investigation into market-specific factors (competition, product fit, service quality) - root cause is not determinable from this dataset alone.

Churn rate by gender

Gender	Customers	Churn Rate
Female	4,543	25.07%
Male	5,457	16.46%

Female customers churn at a meaningfully higher rate than male customers in this dataset. This pattern holds within every geography (France: 20.3% vs 12.7%; Germany: 37.6% vs 27.8%; Spain: 21.2% vs 13.1%), suggesting it is a consistent effect rather than a geography-driven artifact. The underlying cause is not identifiable from these fields alone and would benefit from qualitative follow-up (service experience, product fit by segment).

Churn rate by number of products held

Products Held	Customers	Churn Rate
1	5,084	27.71%
2	4,590	7.58%
3	266	82.71%
4	60	100.00%

Honest caveat: this relationship is sharply non-linear (a U-shape, not a straight line), which is why the simple linear correlation between NumOfProducts and churn is weak and slightly negative (-0.048) despite this being one of the most powerful predictors in the trained model. Customers with 3 or 4 products are a small slice of the base (326 of 10,000) but churn at 83-100% - this is the most extreme signal in the dataset and should be treated as a priority segment for investigation, since it likely reflects either forced/bundled product sign-ups shortly before account closure, or a specific customer segment mis-served by over-bundling.

Churn rate by age group

Age Band	Customers	Churn Rate
18-30	1,968	7.52%
31-40	4,451	12.09%
41-50	2,320	33.97%
51-60	797	56.21%
61-70	331	31.42%
71+	133	8.27%

Churn rises sharply through middle age and peaks in the 51-60 band at 56.21% - nearly 4.5x the rate of the youngest cohort - before falling again in the oldest customers. Age is the single strongest linear correlate of churn in the dataset ($r = 0.285$) and the top feature by model importance.

Churn by account balance

Segment	Customers	Churn Rate
Zero balance	3,617	13.82%
Non-zero balance	6,383	24.08%

Counterintuitively, customers holding a non-zero balance churn at nearly double the rate of zero-balance customers. This runs against the common assumption that higher-balance customers are inherently 'stickier,' and is worth flagging to a banking stakeholder as a non-obvious finding rather than smoothing it over.

Churn by activity status and credit card ownership

Segment	Churn Rate
Active member	14.27%
Inactive member	26.85%
Has credit card	20.18%
No credit card	20.81%

Active-member status shows a clear, usable gap (nearly double the churn rate for inactive members) and is a plausible lever for engagement campaigns. Credit card ownership shows essentially no relationship with churn (20.18% vs 20.81%) and is the weakest signal among the binary flags - the model's own feature importance ranks it last (0.010).

Correlation with target (linear, numeric features only)

Feature	Correlation with Exited
Age	+0.285
Balance	+0.119
EstimatedSalary	+0.012
HasCrCard	-0.007
Tenure	-0.014
CreditScore	-0.027
NumOfProducts	-0.048 (misleading - see non-linear note above)
IsActiveMember	-0.156

Linear correlation is a limited lens here - NumOfProducts is a strong predictor in the trained (non-linear) model despite an almost-zero linear correlation, which is exactly why a tree-based model was an appropriate choice over a purely linear one for this problem.

Customer segmentation (KMeans clustering)

Beyond the individual-level classifier, the project notebook also performs unsupervised segmentation: 8 behavioral features (CreditScore, Age, Tenure, Balance, NumOfProducts, EstimatedSalary, HasCrCard, IsActiveMember) are standardized, then clustered with KMeans. The optimal cluster count (k=4) was selected via elbow method and silhouette score analysis across k=2 to k=10, and the result was visualized in two dimensions via PCA.

Cluster	Profile (avg. values)	Churn Rate
3	NumOfProducts 2.15, Balance ~10,605 (near-zero)	13.46%
0	Active members (100%), have credit cards (100%)	15.52%
2	No credit card (0%), moderate activity (52%)	19.94%
1	Inactive members (0% active), have credit cards (98%)	32.08%

Honest caveat: Cluster 1 - inactive credit-card holders - churns at more than double the rate of Cluster 3, the lowest-risk segment. This is a coarser signal than the individual-level model but is immediately actionable: it identifies a ready-made target list for a retention campaign without requiring per-customer scoring. Silhouette scores across all tested k values were modest (0.12-0.15), indicating the clusters are real but not sharply separated - a caveat worth stating rather than overselling the segmentation's precision.

11. Feature Engineering

The current implementation uses the ten raw customer attributes directly, with two preprocessing transformations rather than newly engineered features:

- Numeric features (CreditScore, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary) are standardized via scikit-learn's StandardScaler.
- Categorical features (Geography, Gender) are one-hot encoded via OneHotEncoder with drop='first' to avoid redundant dummy columns.
- CustomerId and Surname are excluded from the feature set as non-predictive identifiers.

Honest caveat: no derived/engineered features (e.g. balance-to-salary ratio, age bands, tenure-to-age ratio, or an explicit interaction term for the NumOfProducts non-linearity found in EDA) have been built yet. Given the strong non-linear NumOfProducts effect and the age-band pattern surfaced in EDA, engineered features are a concrete, evidence-backed opportunity for the next iteration - not a generic suggestion.

12. Machine Learning / Model Development

Problem type

Binary classification (Exited: 1 = churned, 0 = retained).

Algorithm

A single Random Forest classifier is implemented and deployed. No alternative algorithms (logistic regression, gradient boosting, neural networks, etc.) have been trained or compared in this codebase.

[Information Needed]

Hyperparameters

Parameter	Value
n_estimators	200
max_depth	10
min_samples_split	5
min_samples_leaf	2
random_state	42

These values are fixed in code. No hyperparameter search (grid search, random search, Bayesian optimization) is implemented - the values above reflect a single manually-chosen configuration, not a tuned optimum.

Honest caveat: this project has two documented model iterations, not one, and both are reported rather than merged. The original notebook (bankcustomersegmentation.ipynb) trained an earlier, unconstrained Random Forest - n_estimators=200 only, default max_depth (unlimited), default min_samples_split/leaf, and a non-stratified train/test split - which scored 86% accuracy, 0.70 churn precision, 0.47 churn recall, and 0.8456 AUC. This was later refined into the hyperparameters listed above (added depth/leaf constraints, stratified split) for the deployed train.py/app.py pipeline, trading some recall for a substantial precision gain. The deployed application uses the refined version; the notebook version is documented here for completeness and transparency about the development history.

Training and validation strategy

A single stratified 80/20 train/test split (random_state=42) is used. No k-fold cross-validation is implemented, so the reported metrics reflect performance on one held-out split rather than an averaged estimate across multiple splits - a meaningful caveat for how much confidence to place in the exact metric values.

Explainability

Feature importance is derived from the Random Forest's built-in impurity-based feature_importances_ and surfaced in the deployed app per prediction. SHAP, LIME, partial dependence plots, and per-customer signed explanations are not implemented.

[Information Needed]

13. Evaluation & Results

Test set performance

Metric	Value
Accuracy	86.80%
ROC-AUC	0.864
Precision - Stay	0.87
Recall - Stay	0.98
F1 - Stay	0.92
Precision - Churn	0.83
Recall - Churn	0.44
F1 - Churn	0.58

Confusion matrix (test set, n = 2,000)

	Predicted Stay	Predicted Churn
Actual Stay	1,556 (TN)	37 (FP)
Actual Churn	227 (FN)	180 (TP)

Honest caveat: this is the central limitation of the current model and is stated plainly rather than minimized. Because 79.6% of customers do not churn, a model can reach ~80% accuracy by defaulting toward the majority class; 86.80% is a genuine improvement over that baseline, but the churn-class recall of 0.44 means the model still misses the majority of customers who actually leave (227 of 407 churners in the test set were missed). Precision of 0.83 means the model rarely raises a false alarm when it does flag someone as high risk - so its outputs can be trusted, but its coverage is incomplete.

Overfitting/underfitting diagnostics (learning curves, train-vs-test gap) and cross-validated variance estimates:

[Information Needed]

Top feature importances

Rank	Feature	Importance
1	Age	0.312
2	Number of Products	0.234
3	Balance	0.109
4	Credit Score	0.078
5	Estimated Salary	0.076
6	Active Member Status	0.066
7	Geography - Germany	0.051
8	Tenure	0.040
9	Gender - Male	0.018
10	Has Credit Card	0.010
11	Geography - Spain	0.008

14. Business Insights

- Germany is a disproportionately high-risk market (32.4% churn vs ~16% elsewhere) - worth a dedicated market-level investigation rather than a one-size-fits-all retention strategy.
- Customers holding 3-4 products churn at 83-100%, despite product cross-sell typically being treated as a retention lever - this specific segment (326 customers in this dataset) should be examined for over-bundling or forced sign-up patterns before scaling further cross-sell campaigns.
- The 51-60 age band is the highest-risk age segment (56.2% churn) - likely tied to specific life-stage financial behavior (e.g. retirement planning, competing offers) that merits targeted messaging.
- Inactive members churn at nearly double the rate of active members (26.9% vs 14.3%) - activity/engagement campaigns are a plausible, low-friction retention lever.
- Non-zero-balance customers churn more than zero-balance customers - a genuinely counterintuitive finding that should inform how 'high-value' customers are treated in retention prioritization; balance alone should not be read as a loyalty signal.
- Credit card ownership shows negligible relationship with churn and is not a useful targeting variable on its own.
- KMeans segmentation surfaces a directly targetable group: inactive credit-card holders (Cluster 1) churn at 32.08%, more than double the lowest-risk segment (13.46%) - a ready-made retention campaign target independent of individual model scores.

15. Recommendations

- Prioritize closing the churn-class recall gap (0.44-0.47 depending on model version) before using this model for high-stakes retention budget decisions - via class weighting, threshold tuning, or resampling.
- Investigate the Germany-market and 3-4-product segments qualitatively (support tickets, exit surveys, sales process review) since the data flags the pattern but not the cause.
- Pilot an engagement campaign targeted at inactive members - especially the inactive-credit-card-holder segment identified by clustering - given the size and directional clarity of that effect, as a low-risk first action while model refinement continues.
- Do not treat high account balance as an implicit indicator of low churn risk in manual retention triage, given the counterintuitive direction of that relationship in this data.

16. Challenges

- Class imbalance (20.37% churn) makes naive accuracy a misleading headline metric - required explicitly reporting class-level precision/recall rather than a single aggregate number.
- The NumOfProducts feature has a strongly non-linear relationship with churn, which is invisible to simple linear correlation and required tree-based modeling and grouped EDA to surface.
- Geography and Gender each show real disparities in churn outcomes; using them as model features requires a fairness review before any production deployment that acts on individual predictions.

Other technical, data, or deployment challenges encountered during development (e.g. environment/versioning issues implied by train.py's note about regenerating models 'using the current sklearn version in your venv'):

[Information Needed]

17. Future Improvements

- Improve churn-class recall via class weighting, probability threshold tuning, or resampling (e.g. SMOTE).

- Train and compare a second model family (e.g. gradient boosting) rather than relying on a single algorithm.
- Introduce k-fold cross-validation and a hyperparameter search rather than a single fixed configuration.
- Add SHAP-based, per-customer explainability alongside the existing global feature importances.
- Engineer features suggested directly by the EDA - e.g. an explicit high-product-count flag, age-band buckets, and a balance-to-salary ratio.
- Add automated testing and CI, and a documented licensing decision, for the codebase.

18. Conclusion

ChurnIQ demonstrates a complete, deployed churn-prediction pipeline - from a clean dataset through a trained Random Forest model to a live, explainable, action-oriented application - built and evaluated with transparent, verifiable metrics rather than curated ones. The model provides real, non-trivial predictive signal (86.80% accuracy, 0.864 AUC) and surfaces genuinely useful, sometimes counterintuitive business findings (the Germany effect, the product-count non-linearity, the balance-churn direction). Its principal limitation - 0.44 recall on the churn class - is well-understood, quantified, and has a clear, concrete improvement path rather than being an unknown risk.

19. Appendix

A. Feature list used by the model

CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary.

B. Files in the training/deployment pipeline

- train.py - loads data, builds preprocessing pipeline, trains the Random Forest, evaluates, and exports model artifacts.
- app.py - Streamlit application that loads the exported artifacts and serves live predictions with explainability and action-plan output.
- models/bank_churn_rf_model.pkl, models/bank_preprocessor.pkl - exported model and preprocessor.

C. Additional artifacts

- bankcustomersegmentation.ipynb - the original exploratory notebook (EDA, cleaning checks, correlation analysis, the exploratory Random Forest, and the KMeans segmentation).
- images/missing_values_heatmap.png, feature_histograms.png, boxplots_by_exited.png, correlation_heatmap.png - EDA visualizations exported from the notebook.
- images/confusion_matrix.png, clustering_optimal_k.png, cluster_pca_visualisation.png, churn_rate_by_cluster.png - model evaluation and segmentation visualizations exported from the notebook.
- App screenshots: [Information Needed]

20. References

- Dataset: Kaggle 'Bank Customer Churn' dataset (Churn_Modelling.csv), reformatted as BANK_CUSTOMER_SEGMENTATION.csv.
- scikit-learn documentation (RandomForestClassifier, ColumnTransformer, StandardScaler, OneHotEncoder).

- Streamlit documentation.

Additional academic or industry references consulted during development:

[Information Needed]